

Using tropical methods to compute the degrees of some varieties

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Maggie Lai. Using tropical methods to compute the degrees of some varieties.

(Professor Kalina Mincheva, Mathematics)

It is sometimes easier to see the answer to a mathematical question than to determine it through written computations—for example, consider graphing a polynomial to see its intersections with the x-axis instead of factoring it and finding its roots. The objects of study in algebraic geometry are algebraic varieties, which are the points at which a set of polynomial equations vanish. Many properties of algebraic varieties—such as dimension and, under certain conditions, degree—are preserved after tropicalization, a process that linearizes polynomials and thus allows for the use of a broader range of tools in their study.

Here we report the results from studying the tropicalization of some Cayley-Menger varieties, which appear in rigidity theory and have applications in structural engineering, robotics, and biology. In special cases, Cayley-Menger varieties are hypersurfaces and are defined by a single homogeneous polynomial called the circuit polynomial. Although we typically need to know the generating polynomial to find its tropical variety, we can use the combinatorics of the Cayley-Menger varieties to compute the tropicalization without knowing the defining polynomial. In general, this circuit polynomial is not known, and we use tropicalization to provide a lower bound on its degree.

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1 Background

1.1 Tropicalization

Tropical varieties arise from the tropicalization of classical algebraic varieties, which are the points at which a set of polynomial equations vanish. Let F be some field with a non-archimedean valuation $\nu : F \rightarrow \mathbb{T} = \mathbb{R} \cup \{\infty\}$ such that

1. $\nu(a) = \infty \iff a = 0$
2. $\nu(ab) = \nu(a) + \nu(b)$
3. $\nu(a + b) \geq \min\{\nu(a), \nu(b)\}$ for all $a, b \in F$

For example, the trivial valuation $\nu_0 : K \rightarrow \mathbb{B} = \{0, \infty\}$ is an example of such a valuation and the one that we use throughout this project.

$$\nu(a) = \begin{cases} 0_{\mathbb{B}} = \infty & \text{if } a = 0 \\ 1_{\mathbb{B}} = 0 & \text{if } a \neq 0 \end{cases}$$

Consider a classical polynomial $f(x) = \sum_{\mathbf{u} \in \mathbb{N}^n} c_{\mathbf{u}} x^{\mathbf{u}} \in F[x_1, \dots, x_n]$. Its tropicalization $\text{trop}(f)$ is obtained by making the following replacements in the tropical semiring $(\mathbb{T}, \oplus, \odot)$:

$$x_i \oplus x_j := \min\{x_i, x_j\}$$

$$x_i \otimes x_j := x_i + x_j$$

$$c := \text{val}(c)$$

Thus the tropical polynomial $p = \text{trop}(f)$ is the minimum of a finite set of linear functions [4]. Its corresponding tropical variety is a hypersurface $V(p) = V(\text{trop}(f))$, which is the set of all points at which this minimum is achieved at least twice.

This hypersurface is sometimes called the combinatorial shadow of the classical variety because it is an object from polyhedral geometry. In addition to computing

it as the set of points at which the minimum is achieved at least twice, we can find the tropical variety as the $(n - 1)$ -skeleton of the inner normal fan of the Newton polytope of $f = \sum_{\mathbf{u} \in \mathbb{N}^n} c_{\mathbf{u}} x^{\mathbf{u}}$

1. construct the Newton polytope using the exponents of the variables
2. obtain the polyhedral fan dual to the Newton polynomial;
3. take the $(n - 1)$ -skeleton of the dual fan

The *Newton polytope* is the convex hull of the points $\mathbf{u} = (u_1, u_2, \dots, u_n)$ in $\mathbb{N}^n$ such that $x_1^{u_1} x_2^{u_2} \dots x_n^{u_n}$ appears in f . Essentially, it is a set of connected points, whose coordinates are the powers of the variables in each monomial.

In cases when the valuation is nontrivial, we perform a regular subdivision of the Newton polytope by lifting each of the points of the Newton polytope by a weight from its coefficient valuation and recording the folds that appear on the original Newton polytope. In our case, the valuation ν is trivial, so we completely bypass this step.

A polyhedral cone is the positive hull of a finite set of $\mathbb{R}^n$, which we can view as the positive linear span as a collection of rays: $C = \text{pos}(v_1, \dots, v_r) = \{\sum_{i=1}^r \lambda_i v_i : \lambda_i \geq 0\}$. A face of the cone is determined by a ray in the dual of $\mathbb{R}^n$, some $\mathbf{w} \in (\mathbb{R}^n)^\vee$, where $\text{face}_{\mathbf{w}}(C) = \{x \in C : \mathbf{w} \cdot \mathbf{x} \leq \mathbf{w} \cdot \mathbf{y} \text{ for all } \mathbf{y} \in C\}$. We use these two definitions to build our definition of a polyhedral fan F , which is a collection of polyhedral cones such that

1. for every cone $C \in F$, each face of C is also in F
2. the intersection of any two cones in F is a face of each

The inner normal fan of a polytope is the polyhedral fan that consists of a collection of cones where each cone corresponds to a face of the polytope generated by the inward-pointing normal vectors of each face. Therefore, when we take the *dual* of the Newton polytope, we are really finding its inner normal fan.

Finally, we take the $(n - 1)$ -*skeleton* of the normal fan, which means that we are interested in only the cones of the fan that have dimension at most $n - 1$. From here on out, we refer to dimension $n - d$ as codimension- d or $\text{codim-}d$.

This gives us a new combinatorial approach to determining a polynomial from its tropical variety because we can now work in two “directions”:

1. we can begin from a classical polynomial and find its corresponding tropical variety through steps $(1) \rightarrow (2) \rightarrow (3)$, or
2. we can begin from a tropical hypersurface, find its corresponding complete fan (because the hypersurface is a codimension 1 skeleton), compute a polytope that has this as its normal fan, and find the polynomial defined its vertices by going through steps $(3) \rightarrow (2) \rightarrow (1)$.

This polynomial is not unique because there are infinitely many scaled polytopes that have the same normal fan, but the smallest polytope with positive integer vertices is a lower bound for the degree of the lifted polynomial.

1.1.1 Example of direction 1

Let $f(x_1, x_2) = x_1 + x_2 + 1$. The points $\{(1, 0), (0, 1), (0, 0)\}$ form the Newton polytope $\text{Newt}(f)$, from which we can take the $(n - 1)$ -skeleton of its inner normal fan as the three rays $\{(\frac{1}{0}), (\frac{0}{1}), (\frac{-1}{-1})\}$.

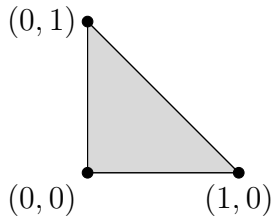


Figure 1: $\text{Newt}(f)$

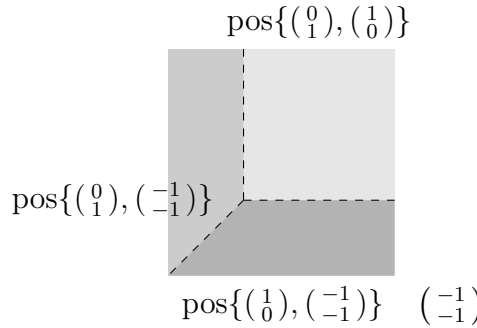


Figure 2: $\text{Fan} \in \mathbb{R}^2$

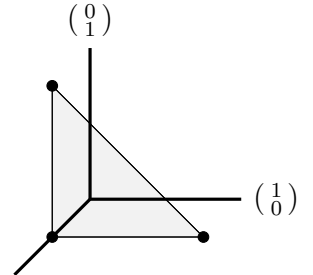


Figure 3: $V(\text{trop}(f))$

1.1.2 Example of direction 2

Suppose we are given the tropical hypersurface whose maximal cones are the rays $\{(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}), (\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}), (\begin{smallmatrix} -1 \\ -1 \end{smallmatrix})\}$. We can easily complete the fan by finding that its full-dimension cones are the positive linear spans of all possible pairs of rays, so we have the cones $\text{pos}\{(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}), (\begin{smallmatrix} 0 \\ 1 \end{smallmatrix})\}$, $\text{pos}\{(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}), (\begin{smallmatrix} -1 \\ -1 \end{smallmatrix})\}$, $\text{pos}\{(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}), (\begin{smallmatrix} -1 \\ -1 \end{smallmatrix})\}$. This is the normal fan of the polytope with points $(0, 0)$, $(0, -1)$, $(1, -1)$, which corresponds to $f(x_1, x_2) = 1 + x_2^{-1} + x_1x_2^{-1}$. Since we specify that our classical polynomials only have positive powers, we multiply $f(x_1, x_2) \cdot x_2 = x_2 + x_1 + 1$, so this is the polynomial with lowest degree corresponding to the tropical hypersurface.

1.2 The Cayley-Menger variety

We aim to apply tropicalization and fan completion to finding a lower bound on the circuit polynomials corresponding to circuits of the algebraic matroid of the Cayley-Menger ideal in $\mathbb{R}^2$.

A matroid $\mathcal{M}$ consists of a finite set E , called the ground set, and a collection $\mathcal{I}$ of subsets of E , called independent sets. Elements of $\mathcal{I}$ satisfy:

1. $\emptyset \in \mathcal{I}$
2. if $I \in \mathcal{I}, J \subset I \implies J \in \mathcal{I}$
3. if $I, J \in \mathcal{I}$ such that $|I| < |J|$, then there exists $e \in J \setminus I$ such that $I \cup \{e\} \in \mathcal{I}$.

A maximal independent set in $\mathcal{I}$ is a basis of $\mathcal{M}$. The subsets of E that are not in $\mathcal{I}$ are called dependent sets, and those that are inclusion-minimal are the circuits of $\mathcal{M}$.

The algebraic matroid of a finite subset E of a finitely generated field extension K/F has the collection of algebraically independent subsets of E as its collection of independent sets $\mathcal{I}$. The support of a polynomial $f \in F[x_e : e \in E]$, denoted $\text{supp}(f)$, is the minimal subset $S \subseteq E$ such that $f \in F[x_e : e \in S]$, i.e. the set of variables appearing in the polynomial. In an algebraic matroid, each circuit C of the matroid corresponds to an irreducible polynomial $p_C \in F[x_e : e \in E]$ with support C that is unique up to multiplication by a constant. These polynomials are called circuit polynomials.

We consider the algebraic matroid of the Cayley-Menger ideal CM_n^d , which comes from distance geometry. Consider n points $p_1, \dots, p_n$ in $\mathbb{R}^d$. We can construct the distance matrix of squared distances between pairs of points, where $x_{i,j} = \|p_i - p_j\|^2$. By adding a new row and column of ones and then zeros on the diagonal, we obtain

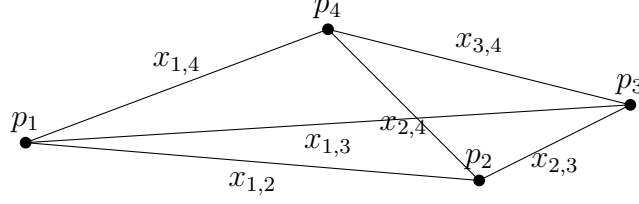


Figure 4: Pairwise distances between n points in $\mathbb{R}^2$

the Cayley matrix:

$$\begin{bmatrix} 0 & 1 & 1 & 1 & \dots & 1 \\ 1 & 0 & x_{1,2} & x_{1,3} & \dots & x_{1,n} \\ 1 & x_{1,2} & 0 & x_{2,3} & \dots & x_{2,n} \\ 1 & x_{1,3} & x_{2,3} & 0 & \dots & x_{3,n} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1,n} & x_{2,n} & x_{3,n} & \dots & 0 \end{bmatrix}$$

Matrix 1: Cayley matrix

When the distances $x_{i,j}$ come from a point set in $\mathbb{R}^d$, the rank of the matrix is at most $d+2$ by Cayley's Theorem, which tells us that each of the $(d+3) \times (d+3)$ submatrices are linearly dependent and have determinant 0. These $(d+3) \times (d+3)$ minors—which are homogeneous polynomials in the indeterminates $x_{i,j}$ —are a generating set for the (n, d) -Cayley-Menger ideal CM_n^d . A variety is the set of points at which a set of polynomial vanishes, and points in the Cayley-Menger variety thus represent squared pairwise distances between n points that can be embedded in $\mathbb{R}^d$, i.e. we can form a valid configuration of n points in $\mathbb{R}^d$ with the given pairwise distances between them.

The algebraic matroid of CM_n^d has the variables $x_{i,j}$ as its ground set, and a subset of these variables S is an independent set if there is no polynomial in the ideal whose support belongs to the set. When we work in $\mathbb{R}^2$, the algebraic matroid $\mathcal{M}(CM_n^2)$ is isomorphic to the $(2, 3)$ -sparsity matroid. This is beneficial because it gives us a much easier way to find the circuits of $\mathcal{M}(CM_n^2)$ and their corresponding circuit polynomials, since there is one for each of the circuits of the 2D rigidity matroid.

The 2D rigidity matroid has the edges of a complete graph K_n as its ground set

and $(2,3)$ -sparse graphs on n vertices are the independent sets. These are graphs where every subgraph $G = (V, E)$ has $|E| \leq 2|V| - 3$. More importantly for our purposes, the circuits of this matroid are easily described: a subgraph $C = (V, E)$ of K_n is a circuit if it satisfies $|E| = 2|V| - 2$ and every subgraph $H = (V_H, E_H)$ satisfies $|E| \leq 2|V| - 3$. For example, both K_4 and the wheel on four vertices, which has a four-cycle and a point connected to each of the cycle vertices, are both circuits in this matroid.

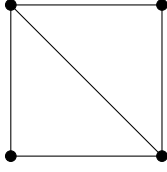


Figure 5: K_4

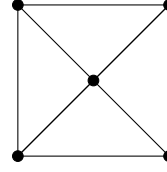


Figure 6: Wheel on 4 vertices

Thus we can rephrase our question regarding the circuit polynomials corresponding to circuits of $\mathcal{M}(CM_n^2)$ to considering these types of graphs, since they are in one-to-one correspondence.

Problem 1 *What is the lower bound of the circuit polynomial of a circuit of $\mathcal{M}(CM_n^d)$ that corresponds to a circuit in the 2D rigidity matroid?*

Since the Cayley-Menger variety gives us information on embeddings of points in Euclidean space, it has many applications in robotics, structural engineering, and protein modeling. Consider the results of nuclear magnetic resonance (NMR), which determines distances between atomic nuclei within molecules. Through a greater understanding of allowable pairwise distances between nuclei in these molecules, we can better describe molecules and, at a larger scale, protein structure.

1.3 Goals and overview of project

We begin to answer this question through the following steps:

1. Choose a rigidity circuit C and obtain the tropicalization of its corresponding Cayley-Menger variety

2. Complete the fan whose $(n - 1)$ -skeleton is the tropical variety
3. Find the polytope of the fan and translate it so that it has only non-negative vertices
4. Reverse the process of taking the Newton polytope, i.e. use the vertices as exponent vectors

The first step is performed using the `tropicalCayleyMenger` function in the rigidity package in `Macaulay2`, and the third step is performed by the `polytope` function (whose input is a complete fan) in the `Polyhedra` package. We define a simple function to reverse the Newton polytope function for the fourth step.

The primary work of this project lies in the second step: obtaining a set of full-dimension cones from the $(n - 1)$ -skeleton whose union is $\mathbb{R}^n$ so that we have a complete fan. This is a process whose applications extend beyond the Cayley-Menger variety, since we seek to complete any fan from its codimension-1 skeleton.

2 Fan Completion

2.1 Computation representation

The tropical variety is the $(n - 1)$ -skeleton of the dual fan, so it is minimally represented as a collection of codim-1 cones (because all intersections between cones are faces and thus belong to the fan), each of which is the positive linear span of some subset of the 1-dimension cones, i.e. rays in $\mathbb{R}^n$. In order to obtain the Newton polytope, we need to get the complete fan corresponding to the $(n - 1)$ -skeleton, i.e. a fan with n -dimension cones whose union is all of $\mathbb{R}^n$ and whose intersections are exactly the $(n - 1)$ -dimension cones.

Problem 2 *What is the complete fan whose n -dimension cones span all of $\mathbb{R}^n$ and whose $(n - 1)$ -skeleton is a specific set of $(n - 1)$ -dimension cones?*

In general, we start with the indexed rays of the fan and the list of cones in the $(n - 1)$ -skeleton. Each cone is represented by the indices of the rays whose positive linear span is the cone. For example, if we have five rays $\{v_1, v_2, v_3, v_4, v_5\}$ in the fan and one codim-1 cone is $\text{pos}(\{v_1, v_3\})$, then it is represented as $\{1, 3\}$. We have two counting checks that provide us a guideline for proceeding:

1. Each codim-1 cone is a face of exactly two full-dimension cones.
2. Each full-dimension cone has at least n codim-1 cones as faces.

2.2 General algorithm

Consider a cone σ of codim-1 in the tropical variety. As stated previously, σ is in two full-dimension cones (and is their intersection) in the complete fan. The cone is the positive linear span of its rays, so points $(x_1, \dots, x_n) \in \sigma$ lie on a hypersurface given by a linear equation $f(x_1, \dots, x_n) = b$. There are points above and below this hypersurface, and these are exactly the points $(x_1, \dots, x_n)$ such that $f(x_1, \dots, x_n) > b$ and $f(x_1, \dots, x_n) < b$ respectively.

In the quotient lattice $\mathbb{Z}^n/N(\sigma)$, where $N(\sigma) :=$ the sublattice generated by the rays of σ , the codim-1 cones adjacent to σ will have image either positive or negative numbers. Because we have all of the codim-2 cones as well, we can determine which codim-1 cones are adjacent to σ by finding those whose intersection with σ belongs to the set of codim-2 cones. Then some subset of the codim-1 cones adjacent to σ with image that is a positive number are faces of the first cone, and some subset of the codim-1 cones adjacent to σ with image that is a negative number are faces of the second cone.

Computationally, assume that a particular codim-1 cone σ is the positive linear span of r rays. We can construct A , the $n \times r$ matrix of its rays. Then the kernel of the column space of A is one-dimensional because σ is of codim-1, so it has a single basis vector u which is normal to σ , i.e. $u \cdot x = 0$ for all $x \in \sigma$. We can apply this u as the projection from the lattice to the quotient lattice:

$$\pi : \mathbb{Z}^n \mapsto \mathbb{Z}^n/N(\sigma)$$

$$x \mapsto u \cdot x$$

so if τ is a codim-1 cone such that $(\sigma \cap \tau) \in$ the set of codim-2 cones, then $\pi(\tau) \neq 0$ and we can use the sign to determine whether it is “above” or “below” σ .

The following algorithm is developed without consideration of the codim-2 cones and counts the incidence of rays appearing in the faces of a potential full-dimension

cone. For any such cone, we expect that there are at least n codim-1 cones acting as its faces, i.e. rays need to appear “often enough” for the full-dimension cone to be fully bounded by codim-1 faces.

Algorithm 1 Find Full-Dimension Cones

```

1: Input: rays, a  $n \times k$  matrix where  $\text{rays}(:, i)$  is the  $i$ th ray, and codim1, a collection
   of sets corresponding to the rays generating each codim-1 cone
2: Output: cones, a collection of sets corresponding to the rays generating each full-
   dimension cone

3: for  $i = 1$  to total number of sets in codim1 do
4:    $r \leftarrow |\text{codim}(i)|$ , the number of rays in the cone codim( $i$ )
5:    $A \leftarrow (n \times r)$  matrix of the rays in codim( $i$ )
6:    $u \leftarrow \ker(A^T)$  the ray perpendicular to codim( $i$ )
7:    $v \leftarrow u^T \cdot \text{rays}$  a  $(1 \times r)$  matrix, where the  $i$ th element tells us whether the  $i$ th ray is
     above (if it is  $> 0$ ) or below ( $< 0$ ) the cone

8:   for subsets  $B$  of the elements of  $v$  that are  $> 0$  do
9:      $L \leftarrow \uplus$  sets corresponding to codim-1 cones that are subsets of  $B$ , a multiset
10:    if at least  $n$  codim-1 cone sets are subsets of  $B$  then
11:       $X_j \leftarrow$  the number of times the ray  $j$  appears in  $L$  for  $j \in \{1, \dots, r\}$ 
12:       $\sigma_B \leftarrow \text{Var}(X)$ 
13:    end if
14:  end for

15:  Add the  $B$  such that  $\sigma_B$  is the smallest to cones as a full-dimension cone

16:  for subsets  $C$  of the elements of  $v$  that are  $< 0$  do
17:    Perform steps 9-13 for the subsets of  $C$ 
18:  end for
19:  Add the  $C$  such that  $\sigma_C$  is the smallest to cones as a full-dimension cone
20: end for

21: return cones

```

A full algorithm and other computations can be accessed in the Github repository at <https://github.com/themaggiest/fan-completion>.

2.3 Example on K_4

2.3.1 Starting from the polynomial

The circuit polynomial for K_4 is known:

$$\begin{aligned}
p_{K_4} = & x_{3,4}x_{1,2}^2 + x_{3,4}^2x_{1,2} + x_{1,3}x_{2,3}x_{1,2} - x_{1,4}x_{2,3}x_{1,2} - x_{1,3}x_{2,4}x_{1,2} + x_{1,4}^2x_{2,3} \\
& + x_{1,3}x_{2,4}^2 + x_{1,4}x_{2,4}x_{1,2} - x_{1,3}x_{3,4}x_{1,2} - x_{1,4}x_{3,4}x_{1,2} + x_{1,3}^2x_{2,4} \\
& + x_{1,4}x_{2,3}^2 - x_{2,3}x_{3,4}x_{1,2} - x_{2,4}x_{3,4}x_{1,2} + x_{2,3}x_{2,4}x_{3,4} - x_{1,3}x_{2,4}x_{3,4} \\
& - x_{1,3}x_{1,4}x_{2,3} - x_{1,3}x_{1,4}x_{2,4} - x_{1,3}x_{2,3}x_{2,4} - x_{1,4}x_{2,3}x_{2,4} + x_{1,3}x_{1,4}x_{3,4} \\
& - x_{1,4}x_{2,3}x_{3,4}
\end{aligned}$$

Polynomial 1: Circuit polynomial for K_4

It is a homogeneous degree three polynomial in six variables corresponding to the six pairwise distances between any four points and has 22 monomials. $Newt(p_{K_4}) = P_{circ}$ is a polytope with volume $5/8$, and its normal fan F_{circ} has the following f -vector:

codim	0	1	2	3	4	5
number of cones	22	75	90	45	10	1

Table 1: f -vector of normal fan to $Newt(p_{K_4})$

Its fan F_{circ} has the following rays, where each ray is a column of the matrix:

$$\begin{array}{c}
\begin{array}{cccccccccc}
\mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} & \mathbf{7} & \mathbf{8} & \mathbf{9}
\end{array} \\
\left[\begin{array}{cccccccccc}
1 & 0 & 0 & 0 & 0 & 2 & 2 & 0 & -1 & -1 \\
0 & 1 & 0 & 0 & 0 & 2 & -1 & 0 & 2 & -1 \\
0 & 0 & 1 & 0 & 0 & -1 & 2 & 0 & 2 & -1 \\
0 & 0 & 0 & 1 & 0 & 2 & -1 & 0 & -1 & 2 \\
0 & 0 & 0 & 0 & 1 & -1 & 2 & 0 & -1 & 2 \\
0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 2 & 2
\end{array} \right]
\end{array}$$

Matrix 2: Rays of normal fan to $Newt(p_{K_4})$

2.3.2 Starting from the variety

Using the rigidity package, we calculate that the tropical variety corresponding to K_4 is in $\mathbb{R}^5$ and contains 75 codim-1 cones, each of which is the positive linear span of four rays. It has ten rays as its codim-4 cones, which are given as columns in the matrix:

$$\begin{array}{c} \begin{array}{cccccccccc} \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} & \mathbf{7} & \mathbf{8} & \mathbf{9} \end{array} \\ \left[\begin{array}{cccccccccc} -1 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 1 \\ 0 & 0 & -1 & 0 & 0 & -1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 & 0 & 0 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 & 1 & 1 & 1 \end{array} \right] \end{array}$$

Matrix 3: Rays of tropical variety in max convention

Since the rigidity package uses maximum instead of minimum, we multiply each of the rays by (-1) to obtain the following rays:

$$\begin{array}{c} \begin{array}{cccccccccc} \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} & \mathbf{7} & \mathbf{8} & \mathbf{9} \end{array} \\ \left[\begin{array}{cccccccccc} 1 & 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & -1 & -1 & -1 \end{array} \right] \end{array}$$

Matrix 4: Rays of tropical variety in min convention

Using the fan completion algorithm, we find 22 full-dimension cones, 12 of which are defined as the positive linear span of five rays, and 10 of which are defined as the positive linear span of six rays. The 12 cones of five rays each have five codim-1 cones

as faces, while the 10 cones of six rays each have nine codim-1 cones as faces. This completed fan, which we now refer to as F_{var} satisfies two checks:

1. The union of the 22 cones is all of $\mathbb{R}^5$, and the fan is indeed complete.
2. We expect any codim-1 cone to belong to exactly two full-dimension cones, so for 75 codim-1 cones, there should be 150 codim-1 cones as faces of the full-dimension cones. From the 12 cones with five faces and 10 cones with nine faces, we do indeed obtain $12 \cdot 5 + 10 \cdot 9 = 150$ codim-1 cones acting as faces of the full-dimension cones.

This gives us the f -vector

codim	0	1	2	3	4	5
number of cones	22	75	90	45	10	1

Table 2: f -vector of completed fan for tropical variety

This complete fan is the normal fan of a polynomial with 22 monomials in five variables. Because some variables are raised to negative powers, we multiply the polynomial by $x_2x_3x_4$ to obtain

$$\begin{aligned}
f_{K_4} = & x_1x_2x_3 + x_2^2x_3 + x_2x_3^2 + x_1^2x_4 + x_1x_2x_4 + x_1x_3x_4 + x_2x_3x_4 + x_1x_4^2 \\
& + x_1x_3x_5 + x_2x_3x_5 + x_1x_4x_5 + x_2x_4x_5 + x_1x_2 + x_2x_3 + x_1x_4 \\
& + x_3x_4 + x_1x_5 + x_2x_5 + x_3x_5 + x_4x_5 + x_5^2 + x_5
\end{aligned}$$

Polynomial 2: Shifted vertices of polytope from tropical variety

The polynomial is of degree 3, and its polytope P_{var} has volume $5/8$.

2.3.3 Comparing the two methods

The polytope P_{var} whose normal fan is the completed tropical variety and $Newt(p_{K_4}) = P_{circ}$ have the same volume, number of vertices, and f -vector, which is

the dual of the fans' vectors in TABLES 2 and 1.

The rays of the fans F_{var} and F_{circ} are also extremely similar—the first five rays in both are $\mathbf{e}_1, \dots, \mathbf{e}_5$. Because the circuit polynomial p_{K_4} is homogeneous, F_{circ} contains the line $\ell = (1, 1, 1, 1, 1, 1)^T$ as its lineality space. Adding $(0, 0, 0, 0, 0, \ell, \ell, -\ell, -2\ell, -2\ell)$ to the rays of F_{circ} , we get

$$\begin{array}{cccccccccc}
 \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} & \mathbf{7} & \mathbf{8} & \mathbf{9} \\
 \left[\begin{array}{ccccccccc}
 1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & -1 \\
 0 & 1 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & -1 \\
 0 & 0 & 1 & 0 & 0 & 0 & 1 & -1 & 0 & -1 \\
 0 & 0 & 0 & 1 & 0 & 1 & 0 & -1 & -1 & 0 \\
 0 & 0 & 0 & 0 & 1 & 0 & 1 & -1 & -1 & 0
 \end{array} \right]
 \end{array}$$

Matrix 5: Rays of F_{circ} after transforming by lineality space

after eliminating the last row of zeros. This fan is isomorphic to F_{var} with the isomorphism

$$\left[\begin{array}{ccccc}
 1 & 0 & 0 & -1 & 0 \\
 0 & 1 & 0 & -1 & 0 \\
 0 & 0 & 0 & -1 & 1 \\
 0 & 0 & 0 & -1 & 0 \\
 0 & 0 & 1 & -1 & 0
 \end{array} \right]$$

Matrix 6: Isomorphism

and thus they are also normal fans of isomorphic polytopes.

The polynomial that we obtain from the transformed fan is

$$\begin{aligned}
p'_{K_4} = & x_1x_2x_4 + x_1x_3x_4 + x_2x_3x_4 + x_3^2x_4 + x_3x_4^2 + x_1x_2x_5 + x_2^2x_5 \\
& + x_1x_3x_5 + x_2x_3x_5 + x_2x_4x_5 + x_3x_4x_5 + x_2x_5^2 + x_1^2 + x_1x_2 \\
& + x_1x_3 + x_2x_3 + x_1x_4 + x_3x_4 + x_1x_5 + x_2x_5 + x_4x_5 + x_1
\end{aligned}$$

Polynomial 3: Vertices of polytope normal to transformed F_{circ}

We can obtain p'_{K_4} from f_{K_4} by the following change of variables: $x_1 \rightarrow x_5 \rightarrow x_3 \rightarrow x_1$.

2.3.4 Relations between cones

The completed fan of the tropical variety is very regular; each five-ray cone has exactly five faces, and each six-ray cone has exactly nine faces, which is not true in general for arbitrary fans. Their intersections are also very regular: each five-ray cone shares all five of its faces only with six-ray cones, while each six-ray cone shares six of its faces with five-ray cones and three of its faces with other six-ray cones.

The following graph shows the full-dimension cones of the complete fan as nodes and codim-1 faces between full-dimension cones as edges. The orange nodes represent five-ray cones, the blue nodes represent six-ray cones, the green edges are faces between six-ray cones, and the orange edges are faces between six- and five-ray cones. The subgraph of six-ray cones and the faces between them is a 3-regular graph on ten nodes and is indeed the Petersen graph.

When applying the fan completion algorithm, we also get extremely regular results. The five faces of each five-ray cone collectively contain each ray exactly four times, while the nine faces of each six-ray cone collectively contain each ray exactly six times.

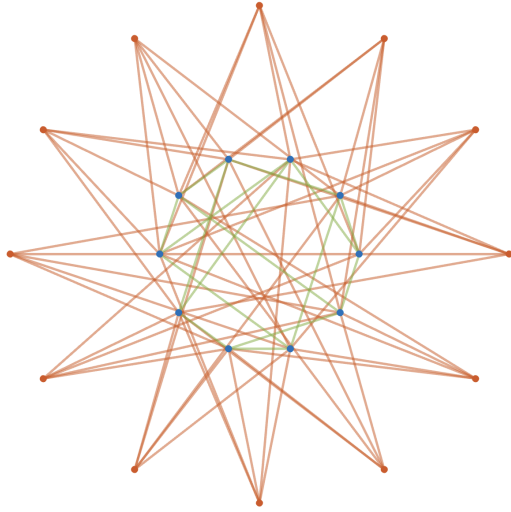


Figure 7: Unlabeled K_4 graph



Figure 8: Labeled K_4 graph

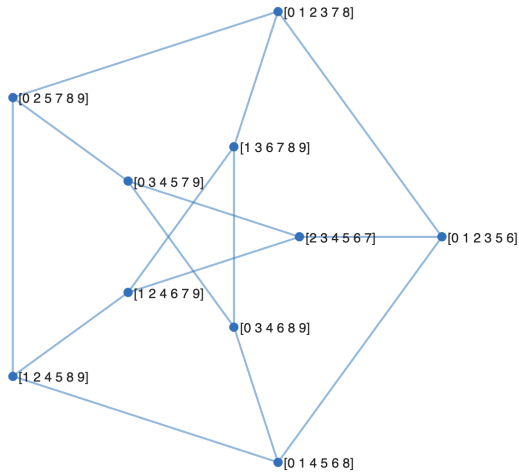


Figure 9: Six-ray cones of K_4 graph

2.3.5 Example of ray containment

Consider the codim-1 cone that consists of rays 3, 4, 5, and 6. The matrix of its rays (taken from MATRIX 3) is

$$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & -1 \\ 0 & -1 & -1 & -1 \end{bmatrix}$$

Matrix 7: Rays of codim-1 cone of variety of K_4

and the basis of its null space is $u = [-1, 0, 1, 0, 0]^T$. When we multiply $u^T M_2$, where M_2 is MATRIX 3, we obtain

$$\begin{array}{cccccccccc} \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} & \mathbf{7} & \mathbf{8} & \mathbf{9} \\ \left[\begin{array}{cccccccccc} 1 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{array} \right] \end{array}$$

which tells us that rays 0 and 8 are to one side of the cone and rays 2 and 7 are to the other side. Without loss of generality, we call rays 0 and 8 “above” the cone and rays 2 and 7 “below” the cone. Note that we could have equivalently used the rays of MATRIX 4 to perform all of these calculations.

Then we only need to consider $\text{pos}(\{3, 4, 5, 6, 0, 8\})$, $\text{pos}(\{3, 4, 5, 6, 0\})$, $\text{pos}(\{3, 4, 5, 6, 8\})$, $\text{pos}(\{3, 4, 5, 6, 2, 7\})$, $\text{pos}(\{3, 4, 5, 6, 2\})$, $\text{pos}(\{3, 4, 5, 6, 7\})$, and $\text{pos}(\{3, 4, 5, 6, 2\})$ as potential full-dimension cones in the complete fan. The first three are the original rays of the codim-1 cone with subsets of the rays above the cone, while the last three are the original rays of the codim-1 cone with subsets of the rays below the cone. For the following calculations, we leave off the pos symbol for ease of notation and discussion.

1. The following codim-1 cones have rays only from $\{3, 4, 5, 6, 0, 8\}$: $\{3, 4, 5, 6\}$,

$\{0, 4, 5, 6\}, \{0, 3, 5, 6\}, \{0, 4, 6, 8\}, \{0, 3, 6, 8\}, \{0, 4, 5, 8\}, \{0, 3, 4, 6\}, \{0, 3, 4, 5\}$

Each ray appears the following number of times:

$$\begin{array}{cccccc} \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} & \mathbf{0} & \mathbf{8} \\ \left[\begin{array}{cccccc} 7 & 5 & 6 & 5 & 6 & 3 \end{array} \right] \end{array}$$

2. The following codim-1 cones have rays only from $\{3, 4, 5, 6, 0\}$: $\{3, 4, 5, 6\}, \{0, 4, 5, 6\}, \{0, 3, 5, 6\}, \{0, 3, 4, 6\}, \{0, 3, 4, 5\}$

Each ray appears the following number of times:

$$\begin{array}{ccccc} \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} & \mathbf{0} \\ \left[\begin{array}{ccccc} 4 & 4 & 4 & 4 & 4 \end{array} \right] \end{array}$$

3. The following codim-1 cones have rays only from $\{3, 4, 5, 6, 0\}$: $\{3, 4, 5, 6\}$

Each ray appears the following number of times:

$$\begin{array}{ccccc} \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} & \mathbf{8} \\ \left[\begin{array}{ccccc} 1 & 1 & 1 & 1 & 1 \end{array} \right] \end{array}$$

Although both $\text{pos}(\{3, 4, 5, 6, 0, 8\})$ and $\text{pos}(\{3, 4, 5, 6, 0\})$ have at least five faces, which is the minimum number of faces for a full-dimension cone, the variance of the number of times that rays occur in $\text{pos}(\{3, 4, 5, 6, 0\})$ is 0 and thus is lower, so we know that, in fact, the codim-1 cone $\text{pos}(\{3, 4, 5, 6\})$ is a face of $\text{pos}(\{3, 4, 5, 6, 0\})$ and not $\text{pos}(\{3, 4, 5, 6, 0, 8\})$ or $\text{pos}(\{3, 4, 5, 6, 8\})$.

We can perform the same calculation for the rays that are “below” the cone to find that there are three codim-1 cones that have rays only from $\{3, 4, 5, 6, 2\}$ and three codim-1 cones that have rays only from $\{3, 4, 5, 6, 7\}$. There are nine codim-1 cones that have rays only from $\{3, 4, 5, 6, 2, 7\}$, and in these cones each of the rays appears a total of six times, giving us our six-ray cone with nine faces. Thus the two full-dimension cones that $\text{pos}(\{3, 4, 5, 6\})$ belongs to are $\text{pos}(\{3, 4, 5, 6, 0\})$ and

$\text{pos}(\{3, 4, 5, 6, 2, 7\})$.

This regularity—exactly four occurrences of each ray in all the faces of a five-ray cone and exactly six occurrences of each ray in all the faces of a six-ray cone—is not always present, and likely has to do with K_4 's regularity as a complete graph.

3 Future Work

Because of the applicability of fan completion to other problems outside of the Cayley-Menger variety, we plan on implementing its algorithm as a function in Macaulay2 for general $(n - 1)$ -skeletons of complete fans.

Specifically in regards to the Cayley-Menger variety, we are continuing to compute the tropicalization of varieties associated to larger circuits in the 2D rigidity matroid, including the wheels on four and five vertices, and the completion of those varieties. For all such circuits on up to six vertices, we will be able to compare our bounds to existing computed circuit polynomials, but we hope to eventually apply our method to larger examples for which there is no existing computation.

4 References

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